

Padmanabhan Vijendran

Big Data Developer / Data Engineer

✉ padmanabhan5789@gmail.com ☎ +1 6475093692 📍 Mississauga, Canada

🌐 <https://www.linkedin.com/in/amazingpaddy/> 🔄 <https://github.com/amazingpaddy>

🔗 <https://amazingpaddy.github.io/>

Profile

- IT professional with **14 years** of experience in developing applications using **Java, Python** and **Big-Data** technologies
- Extensive expertise in working with **Hadoop, Spark 3.x, Spark Structured Streaming, Kafka, Hive** and **HBase**
- **Google Cloud Certified - Professional Data Engineer**
- Strong knowledge of **Python** programming and **Machine learning** concepts
- Proficient in **UNIX commands** and **shell scripting**
- Excellent analytical, communication and monitoring skills
- Adaptable to different technologies and eager to learn new concepts quickly

Professional Experience

Lead Data Engineer, Royal Bank of Canada

06/2023 – present

Project Details: As the lead developer, I spearheaded the design and development of RBC's End-of-Day Contract (EODc) Compliance Checker tool, leveraging **Java 17**. This tool serves various business sectors within RBC, including Retail and Capital Markets. The Compliance Checker automates the validation of daily contract reports, ensuring they meet both industry and internal standards. It flags any discrepancies, providing a robust solution for quality assurance and regulatory compliance across multiple lines of business.

Toronto, Canada

Roles and Responsibilities:

- **Designed and Developed** the EODc Compliance Checker, a Java-based application that automates the validation of complex daily contract reports across multiple business lines.
- **RESTful Web Services:** Leveraged Apache Spring to implement REST services that trigger the EODc report validation process, enhancing the tool's scalability and user interaction.
- Led the migration of application components from Java 8 to Java 17, ensuring compatibility with modern standards and improved performance.
- Transitioned from embedded Tomcat to enterprise-grade Tomcat servers, strengthening application security, reliability, and maintainability.
- Proactively managed and resolved P1/P2 security vulnerabilities, maintaining compliance with organizational and industry standards.
- **Agile Collaboration:** Actively participated in Agile sprints, collaborated with Business Analysts to translate business needs into technical requirements, and ensured timely delivery of high-quality product.
- **Quality Assurance:** Fixed bugs expediently while handling enhancement requests, ensuring optimal application performance and client satisfaction.
- **High-Performance Computing:** Employed multi-threading techniques to process voluminous EODc data packages exceeding 200GB, featuring millions of lines in each file.

- **Code Quality & Testing:** Implemented unit tests with an achievement of over 90% coverage, integrated Sonar Lint for code quality checks, and eliminated code smells.
- **Regression Testing:** Developed functional test cases utilizing the Cucumber Framework to prevent regressions and ensure application stability across releases.
- **DevOps Integration:** Optimized Git workflows and CI/CD pipelines, significantly accelerating the release process and increasing operational efficiency.

Senior BigData Developer, Cognizant Technology Solutions - Client: Ford

08/2021 – 06/2023

Project Details: Ford engineers use reports generated by an **ETL** framework to design future cars with low manufacturing costs. We are migrating the **ETL** framework from **Sqoop** and **Hive** to **Apache Spark Structured Streaming** so that engineers can access near **real-time** reports.

Mississauga, Canada

Roles and Responsibilities:

- Migrated **ETL** framework from **Sqoop/Hive** to **Apache Spark Structured Streaming** for Ford Engineering.
- Developed **Spark Structured Streaming** application using **pyspark** that **reduced data processing time** and increased **report accuracy**.
- Implemented **ETL** pipelines using **Apache Spark, Kafka** and **MongoDB** that enabled engineers to utilize near real-time reports for designing future cars at low manufacturing cost.
- Designed and developed **Java** application to efficiently fetch data from LDAP servers.
- Refactored **Java** applications to improve readability and performance
- Explored new technologies using **POCs** and documented the approaches to implement new project requirements

Senior BigData Developer, Kelly Services - Client: Credit Agricole CIB

03/2018 – 07/2021

Project Details: Cash Management Transactions (CMT)

Singapore

Implemented a high-performance, real-time internal payment engine to orchestrate incoming and outgoing payment transactions. Engineered a resilient, event-driven architecture using **Java 8, Apache Spark Structured Streaming, Kafka**, and **HBase**. Delivered the project following **Agile methodologies** in a dynamic, cross-functional team.

Roles and Responsibilities:

- **Engineered a scalable payment engine** using **Java 8** and **Spark Structured Streaming** to process over **1 million transactions per day** with **sub-second latency**, directly supporting critical business operations.
- **Designed and built a robust data pipeline** with **Kafka** and **HBase** that enabled real-time ingestion, storage, and analysis of transaction data, ensuring data integrity and rapid access for downstream systems.
- **Solved a critical integration challenge** by developing **custom source and sink connectors for IBM MQ** using Spark Structured Streaming. This initiative **improved data reliability and boosted throughput by over 40%** by eliminating a middleware component.
- **Reduced manual effort by 50%** by creating **Python scripts** to automate demo tool execution and performance report generation, allowing the team to focus on core development.
- **Streamlined the CI/CD pipeline** by developing **Jenkins pipelines** with **Groovy** and **shell scripting**, automating the build, test, and deployment of code changes to production.

- **Enhanced code quality and team efficiency** by automating functional tests with the **Cucumber** framework and setting up application monitoring dashboards using the **ELK stack**, providing real-time visibility into performance and system health.
- **Developed a versatile data layer** that enabled seamless communication between the payment engine and various internal applications by translating proprietary formats into a standardized message structure.

Big Data Developer, *Cognizant Technology Solutions - Client: Ford*

02/2016 – 11/2017

Project Details: I worked on the project that generated reports to help Ford Engineer to design a future cars at low manufacturing cost. We implemented **ETL** using **BigData** tools such as **Apache Spark 2.3**, **Hive**, **Sqoop** and **Oozie**.

Chennai, India

Roles and Responsibilities:

- Extracted data from various Oracle databases to **HDFS** and **Hive** using **Sqoop** scripts
- Transformed data to the required schema using **Hive** scripts
- Developed **Java** application to generate Excel reports from **Hive** table data using **Apache POI**
- Explored ETL with **Alteryx**, **Apache Spark** using **POCs**
- Automated manual tasks using **Python** and **shell scripts**

IT Analyst, *Tata Consultancy Services - Client: Target, USA*

01/2011 – 01/2016

Project Details: I worked on the project that processed product images and details for Target website using **ETL Java** application and **SQL** scripts.

Chennai, India

Roles and Responsibilities:

- Developed **ETL Java** application to process product information using **Map-Reduce**
- Wrote **SQL** scripts to transform product related data
- Automated manual tasks using **Python**
- Investigated and analyzed application **support incidents** through to root cause and fix
- Proactively monitored services and responded to alerts raised by customers and **monitoring** systems, ensuring the availability and performance of the **Teamsite** Content Management Systems

Skills

Programming Languages

Java 17, Python 3

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JAVA EE Frameworks

Spring Framework 6 (Core, Dependency Injection, Spring Data JPA, Spring Security), Jakarta EE (CDI, Faces, Weld), PrimeFaces (UI development)

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BigData Tools

Apache Spark, Kafka, Hive, Hadoop

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Databases

MySQL, MongoDB, HBase

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Machine Learning

Algorithms: XGBoost, RandomForest, Logistic

Regression, SVM

ML Python libraries: pandas, numpy, scikit-learn

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Cloud

AWS, Azure, GCP

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Automation Scripts

Shell, Batch, Powershell, Selenium

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DevOps

Docker, Jenkins, Github Actions

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Testing Frameworks

Junit, Cucumber

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Other Tools

Intellij, Eclipse, Git, Gitlab, Putty, ServiceNow, Rally.

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Certificates

- Google Cloud Certified - Professional Data Engineer [↗](#)
- Microsoft Certified: Azure AI Fundamentals [↗](#)
- AWS Cloud Practitioner [↗](#)

Courses

GCP Data Engineering, *Google Qwik-Labs* [↗](#)

Education

Bachelor of Engineering - Electronics and Communication, *Anna University* 2006 – 2010
Chennai, India